MATH 2060 Homework 13

Zhengyu Zou

December 12, 2025

Collaborators: Ethan Bove, Roberta Pagliaro, Anamaria Perez

FOAG 25.1.G.

Solution:

(a) We first show that $\pi_*\Omega_{C/S}^{\otimes n}$ is a vector bundle of rank $(2n-1)(g-1)$. Let $q \in S$ be an arbitrary point. Consider the map $\Phi_q^1 : R^1\pi_*\Omega_{C/S}^{\otimes n} \to H^1(\Omega_{C/S}^{\otimes n}|_{C_p})$. Notice that $\Omega_{C/S}|_{C_p} \cong \omega_{C_p}$ is the dualizing sheaf, so by Serre duality and the fact that $\deg \omega_{C_p}^{\otimes n} = n(2g-2) > 2g-2$, we must have $h^1(\omega_{C_p}^{\otimes n}) = 0$. It follows that Φ_q^1 is surjective.

Now, from the Cohomology and Base Change Theorem (CBCT), we see that Φ_q^1 is an isomorphism for all $q \in S$. This in particular implies $R^1\pi_*\Omega_{C/S}^{\otimes n}|_q = 0$, so by the second part of CBCT, we see that the map Φ_q^0 is an isomorphism, so $\pi_*\Omega_{C/S}^{\otimes n}|_q$ is isomorphic to $H^0(\omega_{C_p}^{\otimes n})$. Notice that the latter has rank

$$h^0(\omega_{C_p}^{\otimes n}) = \deg \omega_{C_p}^{\otimes n} + 1 - g = n(2g-2) + 1 - g = (2n-1)(g-1).$$

Therefore, $\pi_*\Omega_{C/S}^{\otimes n}|_q$ must have the same rank, and in particular since Φ_q^{-1} is surjective, the second part of CBCT implies $\pi_*\Omega_{C/S}^{\otimes n}$ is locally free of rank $(2n-1)(g-1)$.

Next, to see that the push-pull map of any morphism $\psi : S' \to S$ is an isomorphism, we consider the following Cartesian diagram:

$$\begin{tikzcd} C' \arrow[r, "\rho'"] \arrow[d, "\pi'"'] & C \arrow[d, "\pi"] \\ S' \arrow[r, "\rho"'] & S \end{tikzcd}$$

We know that the cotangent bundle commutes with base change, so $(\rho')^*\Omega_{C/S} \cong \Omega_{C'/S'}$. The push-pull map

$$\Phi_{S'}^0 : \rho^*\pi_*\Omega_{C/S} \longrightarrow \pi'_*(\rho')^*\Omega_{C/S} \cong \pi'_*\Omega_{C'/S'}$$

is an isomorphism.

(b) If S is reduced, we may invoke Grauert's Theorem. It suffices to show that $h^0(C_p, \Omega_{C/S}|_{C_p})$ is locally constant and equals g. From the argument in part (a), we see that $\Omega_{C/S}|_{C_p} \cong \omega_{C_p}$, and we know that $h^0(C_p, \omega_{C_p}) = g$. Then, Grauert's Theorem tells us $\pi_*\Omega_{C/S}$ is locally free

of rank g. The commutativity of base change follows from the same argument as part (a), so we will omit it.

$\square$

FOAG 25.1.I.

Solution: We first show that $\pi_*\mathcal{O}_X$ is a line bundle, and the map $\mathcal{O}_Y \to \pi_*\mathcal{O}_X$ is surjective on fibers. To see the former, for each point $q \in X$, we may consider $\Phi_q^0 : \pi_*\mathcal{O}_X|_q \to H^0(\mathcal{O}_{X_q})$. Notice that the latter ring is a ring of constant functions on the fiber $X_q \to q$, which is isomorphic to $\kappa(q)$. The composed map

$$\mathcal{O}_Y \otimes \kappa(q) \to \pi_*\mathcal{O}_X \otimes \kappa(q) \to H^0(\mathcal{O}_{X_q}) \cong \kappa(q)$$

maps the value of a function on Y at the point q to the constant function on X_q that takes that specific value, so it is in particular surjective. It follows that the second map, which is precisely Φ_q^0, must also be surjective. By CBCT, Φ_q^0 is an isomorphism, so the map $\mathcal{O}_Y|_q \to \pi_*\mathcal{O}_X|_q$ is an isomorphism, which implies $\pi_*\mathcal{O}_X|_q$ is a rank-1 vector space. In particular, CBCT implies $\pi_*\mathcal{O}_X$ is a line bundle. In particular, since the map $\pi_*\mathcal{O}_X \otimes \kappa(q) \to H^0(\mathcal{O}_{X_q})$ is an isomorphism, we know that $\mathcal{O}_Y \otimes \kappa(q) \to \pi_*\mathcal{O}_X \otimes \kappa(q)$ must be surjective, so the map between line bundles $\mathcal{O}_Y \to \pi_*\mathcal{O}_X$ is surjective on the fibers. We conclude that this map of line bundles must locally be multiplication by a unit, so it's an isomorphism. $\square$

FOAG 25.3.A.

Solution: We start with a natural isomorphism of functors $\operatorname{Hilb}_{\mathbb{Z}} \mathbb{P}^n \to \operatorname{Mor}(-, \operatorname{Hilb}_{\mathbb{Z}} \mathbb{P}^n)$. Then, the identity map $\operatorname{Hilb}_{\mathbb{Z}} \mathbb{P}^n \to \operatorname{Hilb}_{\mathbb{Z}} \mathbb{P}^n$ corresponds uniquely to a closed subscheme $Z \hookrightarrow \operatorname{Hilb}_{\mathbb{Z}} \mathbb{P}^n \times \mathbb{P}^n_{\mathbb{Z}}$, with the following diagram:

$$\begin{tikzcd} Z \arrow[r, hook, "\text{closed}"] \arrow[d, hook, "\text{flat}"'] & \operatorname{Hilb}_{\mathbb{Z}} \mathbb{P}^n \times \mathbb{P}^n_{\mathbb{Z}} \\ \operatorname{Hilb}_{\mathbb{Z}} \mathbb{P}^n & \end{tikzcd}$$

Now, the morphism $\pi : B \to \operatorname{Hilb}_{\mathbb{Z}} \mathbb{P}^n$ gets mapped to the morphism $\pi^* : \operatorname{Mor}(\operatorname{Hilb}_{\mathbb{Z}} \mathbb{P}^n, \operatorname{Hilb}_{\mathbb{Z}} \mathbb{P}^n) \to \operatorname{Mor}(B, \operatorname{Hilb}_{\mathbb{Z}} \mathbb{P}^n)$ by precomposing with the map π. The map $\pi = \pi^* \operatorname{Id}$ (here the identity is over $\operatorname{Hilb}_{\mathbb{Z}} \mathbb{P}^n$) then corresponds under the natural isomorphism $\operatorname{Hilb}_{\mathbb{Z}} \mathbb{P}^n \to \operatorname{Mor}(-, \operatorname{Hilb}_{\mathbb{Z}} \mathbb{P}^n)$ to the pullback

$$\begin{tikzcd} Y \arrow[r, hook, "\text{closed}"] \arrow[d, hook, "\text{flat}"'] \arrow[dr] & \mathbb{P}^n_B \arrow[dr] & \\ B \arrow[dr] & Z \arrow[r, hook, "\text{closed}"] \arrow[d, hook, "\text{flat}"'] & \operatorname{Hilb}_{\mathbb{Z}} \mathbb{P}^n \times \mathbb{P}^n_{\mathbb{Z}} \\ & \operatorname{Hilb}_{\mathbb{Z}} \mathbb{P}^n & \end{tikzcd}$$

This completes the proof. $\square$

FOAG 25.3.G.

Solution: Let F be the map (the candidate natural bijection) that takes a rank-k quotient $\mathcal{O}_B^{\oplus n} \to \mathcal{Q}$ to the flat family

$$\begin{tikzcd} \mathit{Proj}_B \operatorname{Sym}^\bullet \mathcal{Q} \arrow[r, hook, "\text{closed}"] \arrow[d, "\text{flat, f. pr.}"'] & \mathbb{P}_B^{n-1} \arrow[dl, "\pi"] \\ B & \end{tikzcd}$$

Let G be the map that takes a flat family $X \hookrightarrow \mathbb{P}_B^{n-1}$ over B to the rank-k quotient $\mathcal{O}_B^{\oplus n} \to \pi_* \mathcal{O}_X$.

We would like to show that $G \circ F$ and $F \circ G$ are both identities. We start by showing $G \circ F$ is the identity. Given a rank-k quotient $\mathcal{O}_B^{\oplus n} \to \mathcal{Q}$ over B we would like to show that applying $G \circ F$ to $\mathcal{Q}$ retrieves the same map. Denote by $X' = \mathit{Proj}_B \operatorname{Sym}^\bullet \mathcal{Q}$, which is the closed subscheme we would obtain by applying F to $\mathcal{Q}$. More concretely, we have a diagram

$$\begin{tikzcd} X' = \mathit{Proj}_B \operatorname{Sym}^\bullet \mathcal{Q} \arrow[r, hook, "\text{closed}"] \arrow[d, "\text{flat, f. pr.}"'] & \mathit{Proj}_B \operatorname{Sym}^\bullet \mathcal{O}_B^{\oplus n} \arrow[dl, "\pi"] \\ B & \end{tikzcd}$$

From the construction of global proj, we know that the sheaf $\pi_* \mathcal{O}_{X'}(1)$ is the same as the sheaf $\mathcal{Q}$ over B, and that the closed embedding $X' \hookrightarrow \mathbb{P}_B^{n-1}$ is induced by the map of graded coherent sheaves $\operatorname{Sym}^\bullet \mathcal{O}_B^{\oplus n} \to \operatorname{Sym}^\bullet \mathcal{Q}$, where the first graded piece is given by $\mathcal{O}_B^{\oplus n} \to \mathcal{Q}$, but that's precisely the map $\pi_* \mathcal{O}_{\mathbb{P}_B^{n-1}}(1) \to \pi_* \mathcal{O}_{X'}(1)$. This shows that $G \circ F$ is the identity.

Next, we show that $F \circ G$ is the identity. We would like to show that X is isomorphic to $\mathit{Proj}_B \operatorname{Sym}^\bullet \pi_* \mathcal{O}_X(1)$, and the image of the embedding $\mathit{Proj}_B \operatorname{Sym}^\bullet \pi_* \mathcal{O}_X(1) \hookrightarrow \mathbb{P}_B^{n-1}$ is precisely X. To see this, we show that both closed embeddings have the same ideal sheaf. The first closed embedding is

$$X'' = \mathit{Proj}_B \operatorname{Sym}^\bullet \pi_* \mathcal{O}_X(1) \hookrightarrow \mathit{Proj}_B \operatorname{Sym}^\bullet \mathcal{O}_B^{\oplus n},$$

which corresponds to a map of graded sheaves $\operatorname{Sym}^\bullet \mathcal{O}_B^{\oplus n} \to \operatorname{Sym}^\bullet \pi_* \mathcal{O}_X(1)$. The first graded piece gives us a map $\mathcal{O}_B^{\oplus n} = \pi_* \mathcal{O}_{\mathbb{P}_B^{n-1}}(1) \to \pi_* \mathcal{O}_X(1)$.

To understand the ideal sheaf $\mathcal{I}_{\mathbb{P}_B^{n-1}/X''}$ we need to understand the structure sheaf of X''. We claim that the structure sheaf of X'' is precisely $\mathcal{O}_X$. This is because $\pi_* \mathcal{O}_{X''}(1)$ is the first graded piece of $\operatorname{Sym}^\bullet \pi_* \mathcal{O}_X(1)$ by construction of global proj, but that's simply $\pi_* \mathcal{O}_X(1)$. It follows that $\pi_* \mathcal{O}_X(1) \cong \pi_* \mathcal{O}_{X''}(1)$. $\square$

FOAG 25.3.H.

Solution: Let p_1 and p_2 be projections defined below:

$$\begin{tikzcd} \mathbb{P}^n \times \mathbb{P}^{\binom{n+d}{d}-1} \arrow[r, "p_2"] \arrow[d, "p_1"'] & \mathbb{P}^{\binom{n+d}{d}-1} \arrow[d] \\ \mathbb{P}^n \arrow[r] & \mathbb{Z} \end{tikzcd}$$

Consider the line bundle $\mathcal{L}$ on $\mathbb{P}^n \times \mathbb{P}^{\binom{n+d}{d}-1}$ given by $p_1^*\mathcal{O}(d) \otimes p_2^*\mathcal{O}(1)$. Let f be a global section given by

$$\sum_{|I|=d} a_I x^I,$$

where I is a multi-index over $0, 1, \ldots, n$, and $x^I = \prod_{i \in I} x_i$. We let $\mathcal{H}$ be the vanishing locus of f. Denote by $i : \mathcal{H} \hookrightarrow \mathbb{P}^n \times \mathbb{P}^{\binom{n+d}{d}-1}$ the closed embedding. We would like to show that $p_2 \circ i$ is flat and finitely presented.

Clearly, p_2 is flat. Also, on the fibers of $i \circ p_2$, the section f is then a degree-d homogeneous polynomial over $x_0, \ldots, x_n$, which is a nonzero-divisor on the fiber $\mathbb{P}^n$. By the slicing criterion on the source, the composition $i \circ p_2$ must also be flat. Finally, we can treat p_2 as the projection

$$\mathbb{P}^n \times \mathbb{P}^{\binom{n+d}{d}-1} \cong \mathit{Proj}_{\mathbb{P}^{\binom{n+d}{d}-1}} \operatorname{Sym}^\bullet \mathcal{O}^{\oplus(n+1)}_{\mathbb{P}^{\binom{n+d}{d}-1}} \longrightarrow \mathbb{P}^{\binom{n+d}{d}-1}.$$

This map is clearly finitely presented, since the sheaf $\mathcal{O}^{\oplus(n+1)}$ is finite-type over $\mathbb{P}^{\binom{n+d}{d}-1}$. We conclude that $i \circ p_2$ is finitely presented. $\square$

FOAG 25.3.I.

Solution: Let $b \in B$ be an arbitrary point. We first observe that $H^i(\mathbb{P}^n_{\kappa(b)}, \mathcal{O}_{\mathbb{P}^n_{\kappa(b)}}(d)) = 0$ for all $i > 0$ by the classification of cohomology groups of $\mathcal{O}_{\mathbb{P}^n}(m)$ over $\mathbb{P}^n$. By CBCT, we see that $R^i\pi_*\mathcal{O}_{\mathbb{P}^n_B}(d)|_b = 0$ for all $b \in B$, so in particular $R^i\pi_*\mathcal{O}_{\mathbb{P}^n_B}(d) = 0$ for all $i > 0$.

Next, we know that X_b is a degree-d hypersurface in $\mathbb{P}^n_{\kappa(b)}$. We then have an exact sequenece

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^n_{\kappa(b)}} \longrightarrow \mathcal{O}_{\mathbb{P}^n_{\kappa(b)}}(d) \longrightarrow \mathcal{O}_{X_b}(d) \longrightarrow 0,$$

which induces a long exact sequence with pieces

$$\cdots \longrightarrow H^i(\mathbb{P}^n_{\kappa(b)}, \mathcal{O}_{\mathbb{P}^n_{\kappa(b)}}(d)) \longrightarrow H^i(X_b, \mathcal{O}_{X_b}(d)) \longrightarrow H^{i+1}(\mathbb{P}^n_{\kappa(b)}, \mathcal{O}_{\mathbb{P}^n_{\kappa(b)}}) \longrightarrow \cdots.$$

Notice that for $i > 0$, the first and the third cohomology groups are both 0, so the middle one must also be 0. Again, by CBCT, $R^i\pi_*\mathcal{O}_X(d)|_b = 0$ for all $b \in B$, so in particular $R^i\pi_I\mathcal{O}_X(d) = 0$ for all $i > 0$. Moreover, the long exact sequence of cohomology groups

$$\cdots \longrightarrow H^i(X_b, \mathcal{O}_{X_b}(d)) \longrightarrow H^{i+1}(\mathbb{P}^n_{\kappa(b)}, \mathcal{I}_X(d)) \longrightarrow H^{i+1}(\mathbb{P}^n_{\kappa(b)}, \mathcal{O}_{\mathbb{P}^n_{\kappa(b)}}(d)) \longrightarrow \cdots$$

implies $H^{i+1}(\mathbb{P}^n_{\kappa(b)}, \mathcal{I}_X(d)) = 0$ for all $i > 0$, so $R^{i+1}\pi_*\mathcal{I}_X(d) = 0$ for $i > 0$ by CBCT.

We would like to show that $R^1\pi_*\mathcal{I}_X(d) = 0$. To see this, consider the following part of the long exact sequence

$$\cdots \longrightarrow H^0(\mathbb{P}^n_{\kappa(b)}, \mathcal{O}_{\mathbb{P}^n_{\kappa(b)}}(d)) \longrightarrow H^0(X_b, \mathcal{O}_{X_b}(d)) \longrightarrow H^1(\mathbb{P}^n_{\kappa(b)}, \mathcal{I}_X(d)) \longrightarrow 0$$

The map $H^0(\mathbb{P}^n_{\kappa(b)}, \mathcal{O}_{\mathbb{P}^n_{\kappa(b)}}(d)) \to H^0(X_b, \mathcal{O}_{X_b}(d))$ is simply restricting global functions on $\mathbb{P}^n$ to a hypersurface, but since both schemes only admit constant global functions, the map is an isomorphism, so we have $H^1(\mathbb{P}^n_{\kappa(b)}, \mathcal{I}_X(d)) = 0$. By CBCT, $R^1\pi_*\mathcal{I}_X(d) = 0$. In particular, the following short exact sequence

$$0 \longrightarrow \pi_*\mathcal{I}_X(d) \longrightarrow \pi_*\mathcal{O}_{\mathbb{P}^n_B}(d) \longrightarrow \pi_*\mathcal{O}_X(d) \longrightarrow 0$$

is a map of vector bundles.

Finally, to compute the ranks of the three vector bundles, it suffices to compute them on the fibers. By CBCT, the fiber $\pi_*\mathcal{O}_{\mathbb{P}^n_B}(d)|_b$ at $b \in B$ is precisely $H^0(\mathbb{P}^n_{\kappa(b)}, \mathcal{O}_{\mathbb{P}^n_{\kappa(b)}}(d))$, so the rank given by $h^0(\mathbb{P}^n_{\kappa(b)}, \mathcal{O}_{\mathbb{P}^n_{\kappa(b)}}(d)) = \binom{n+d}{d}$. Similarly, the rank of $\pi_*\mathcal{O}_X(d)$ is given by $h^0(X_b, \mathcal{O}_{X_b}(d))$. To compute the latter, we consider again the long exact sequence on cohomology:

$$0 \longrightarrow H^0(\mathcal{O}_{\mathbb{P}^n}) \longrightarrow H^0(\mathcal{O}_{\mathbb{P}^n}(d)) \longrightarrow H^0(\mathcal{O}_{X_b}(d)) \longrightarrow H^1(\mathcal{O}_{\mathbb{P}^n}) = 0.$$

It follows that $h^0(\mathcal{O}_{X_b}(d)) = h^0(\mathcal{O}_{\mathbb{P}^n}(d)) - h^0(\mathcal{O}_{\mathbb{P}^n}) = \binom{n+d}{d} - 1$. Finally, the rank of $\pi_*\mathcal{I}_X(d)$ is given by $\operatorname{rk}\pi_*\mathcal{O}_{\mathbb{P}^n_B}(d) - \operatorname{rk}\pi_*\mathcal{O}_X(d) = 1$. This completes the proof. $\square$

FOAG 25.3.J.

Solution: Similar to 25.3.G, let F be the map that takes a map $B \to \mathbb{P}^{\binom{n+d}{d}-1}$ to the degree-d hypersurface obtained from pulling back the universal hypersurface $\mathcal{H} \to \mathbb{P}^{\binom{n+d}{d}-1}$. Let G be the map that takes a degree-d hypersurface $X \hookrightarrow \mathbb{P}^n_B$ and outputs the map $B \to \mathbb{P}^{\binom{n+d}{d}-1}$ given by the dual of the injection map $\pi_*\mathcal{I}_X(d) \hookrightarrow \mathcal{O}_B^{\oplus(n+1)}$, where $\pi : \mathbb{P}^n_B \to B$ is the canonical projection map.

We would like to show that $G \circ F$ and $F \circ G$ are both identities. We start by showing $G \circ F$ is the identity. A map $B \to \mathbb{P}^{\binom{n+d}{d}-1}$ corresponds canonically to a line bundle $\mathcal{L}$ over B along with a basepoint free surjection $\mathcal{O}_B^{\oplus\binom{n+d}{d}} \twoheadrightarrow \mathcal{L}$. Let $\mathcal{H}_B$ be the pullback

$$\begin{tikzcd}
\mathcal{H}_B \arrow[r, hook] \arrow[d] \arrow[dr] & \mathbb{P}^n_B \arrow[dr] & \\
B \arrow[dr] & \mathcal{H} \arrow[r, hook] \arrow[d] & \mathbb{P}^n \times \mathbb{P}^{\binom{n+d}{d}-1} \\
& \mathbb{P}^{\binom{n+d}{d}-1} &
\end{tikzcd}$$

Denote by f_I the image of $a_I \in \mathcal{O}_B^{\oplus\binom{n+d}{d}}$ under the surjection $\mathcal{O}_B^{\oplus\binom{n+d}{d}} \twoheadrightarrow \mathcal{L}$, where I is a multi-index of size d from the set $\{0, 1, \ldots, n\}$. We see that $\mathcal{H}_B$ is cut out by the vanishing of the section

$$\sum_{|I|=d} f_I x^I$$

from the line bundle $p_1^*\mathcal{L} \otimes p_2^*\mathcal{O}_{\mathbb{P}^n}(d)$. Here, p_1 and p_2 are the projection maps from $\mathbb{P}^n_B \cong B \times \mathbb{P}^n$ to the factor B and $\mathbb{P}^n$ respectively. In particular, by the interpretation of the map $\pi_*\mathcal{I}_{\mathcal{H}_B}(d) \hookrightarrow \pi_*\mathcal{O}_{\mathbb{P}^n_B}(d) \cong \mathcal{O}_B^{\oplus\binom{n+d}{d}}$ as the homogeneous degree-d polynomial that cuts out $\mathcal{H}_B$, we see that this map parametrizes precisely the section $\sum_{|I|=d} f_I x^I$, so taking its dual we recover the original map $B \to \mathbb{P}^{\binom{n+d}{d}-1}$. This proves $G \circ F = \mathrm{Id}$.

To show $F \circ G = \mathrm{Id}$, we are given a closed embedding $X \hookrightarrow \mathbb{P}^n_B$ flat over B such that the fibers over $b \in B$ are degree-d hypersurfaces in $\mathbb{P}^n_{\kappa(b)}$. This gives us a map $\pi_*\mathcal{I}_X(d) \hookrightarrow \mathcal{O}_B^{\oplus\binom{n+d}{d}}$ that corresponds to a section $\sum_{|I|=d} f_I x^I$ that pulls back to the degree-d homogeneous polynomial over $\kappa(b)$ that cuts out $X_b \hookrightarrow \mathbb{P}^n_{\kappa(b)}$. Then, the map $B \to \mathbb{P}^{\binom{n+d}{d}-1}$ induced by the dual of the above map of vector bundles captures precisely this information, so when we pull back the universal hypersurface $\mathcal{H} \to \mathbb{P}^{\binom{n+d}{d}-1}$, we get the closed embedding $\mathcal{H}_B \hookrightarrow \mathbb{P}^n_B$ whose ideal sheaf is the same as the ideal sheaf of X. $\square$

Problem 8

Recall that any vector bundle on $\mathbb{P}^1$ is isomorphic to $\bigoplus \mathcal{O}_{\mathbb{P}^1}(e_i)$ for integers e_i that are unique up to reordering. We say that a vector bundle over $\mathbb{P}^1$ is *balanced* if $|e_i - e_j| \leq 1$ for all i, j.

Show that balancedness is an open property. In other words, let $\mathcal{E}$ be a vector bundle on $B \times \mathbb{P}^1$ (a "family of vector bundles on $\mathbb{P}^1$"). Then show $\{b \in B \mid \mathcal{E}|_{b\times\mathbb{P}^1} \text{ is balanced}\}$ is an open subset of B.

Solution: Let $b \in B$ be a point on which $\mathcal{E}|_{\mathbb{P}^1_b} \cong \mathcal{O}_{\mathbb{P}^1_b}(e)^{\oplus m} \oplus \mathcal{O}_{\mathbb{P}^1_b}(e+1)^{\oplus n}$ for some $e \in \mathbb{Z}$ and $m, n \in \mathbb{Z}_{\geq 0}$. We will show that there exists a neighborhood $U \ni b$ such that for all $b' \in U$, $\mathcal{E}|_{\mathbb{P}^1_{b'}} \cong \mathcal{O}_{\mathbb{P}^1_{b'}}(e)^{\oplus m'} \oplus \mathcal{O}_{\mathbb{P}^1_{b'}}(e+1)^{\oplus n'}$ for some $m', n' \in \mathbb{Z}_{\geq 0}$.

Denote by $\pi : B \times \mathbb{P}^1 \to B$ the projection map to the first factor, and $p : B \times \mathbb{P}^1 \to \mathbb{P}^1$ the projection map to the second factor. We first $\mathcal{E}$ by $p^*\mathcal{O}_{\mathbb{P}^1}(-e-1)$. This gives us $\mathcal{E}|_{\mathbb{P}^1_b} \cong \mathcal{O}_{\mathbb{P}^1_b}(-1)^{\oplus m} \oplus \mathcal{O}_{\mathbb{P}^1_b}^{\oplus n}$. Notice that H^1 vanishes on $\mathbb{P}^1_b$, so Φ^1_b is surjective, and by CBCT we see that Φ^1_b is an isomorphism. In particular, $(R^1\pi_*\mathcal{E})|_U = 0$ in a neighborhood U of b, and $h^1(\mathcal{E}|_{\mathbb{P}^1_{b'}}) = 0$ for all $b' \in U$. This implies $\mathcal{E}|_{\mathbb{P}^1_{b'}}$ cannot have a factor of $\mathcal{O}(e')$ for $e' < -1$, because otherwise we would pick up some positive rank on h^1. This is equivalent to saying that the original $\mathcal{E}$ does not admit a factor of $\mathcal{O}(e')$ for $e' < e$ on the fiber of any $b' \in U$.

Next, we twist $\mathcal{E}$ by $p^*\mathcal{O}_{\mathbb{P}^1}(-e-2)$, so $\mathcal{E}|_{\mathbb{P}^1_b} \cong \mathcal{O}_{\mathbb{P}^1_b}(-2)^{\oplus m} \oplus \mathcal{O}_{\mathbb{P}^1_b}(-1)^{\oplus n}$. Notice that H^0 vanishes on $\mathbb{P}^1_b$, so Φ^0_b is surjective, and by CBCT we see that Φ^1_b is an isomorphism. In particular, $(\pi_*\mathcal{E})|_V = 0$ in a neighborhood V of b, and $h^0(\mathcal{E}|_{\mathbb{P}^1_{b'}}) = 0$ for all $b' \in V$. This implies $\mathcal{E}|_{\mathbb{P}^1_{b'}}$ cannot have a factor of $\mathcal{O}(e'')$ for $e'' > -1$, because otherwise we would pick up some positive rank on h^0. This is equivalent to saying that the original $\mathcal{E}$ does not admit a factor of $\mathcal{O}(e'')$ for $e'' > e+1$ on the fiber of any $b' \in V$.

We conclude that $U \cap V$ is an open neighborhood of b on which $\mathcal{E}$ is balanced. $\square$